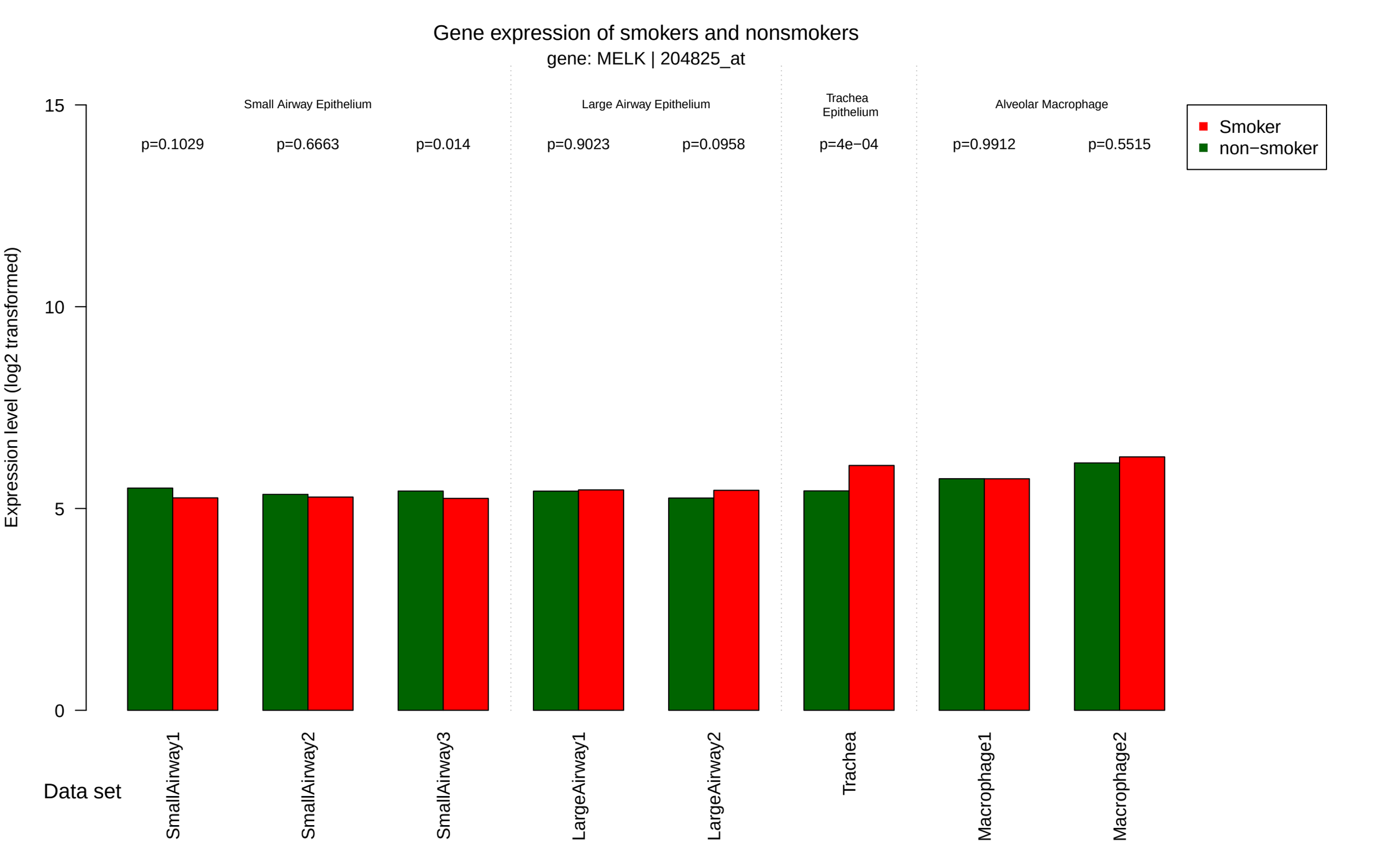
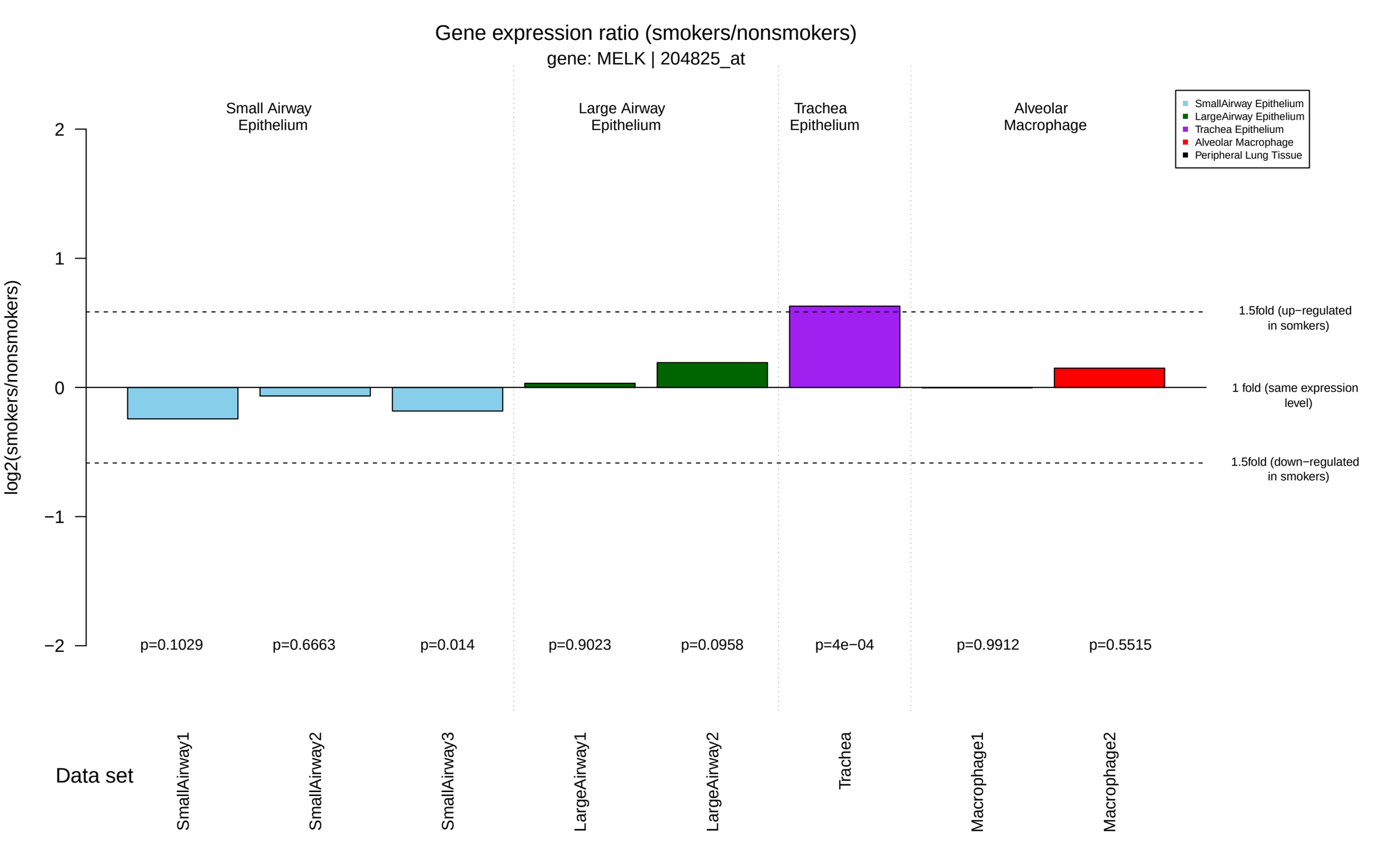
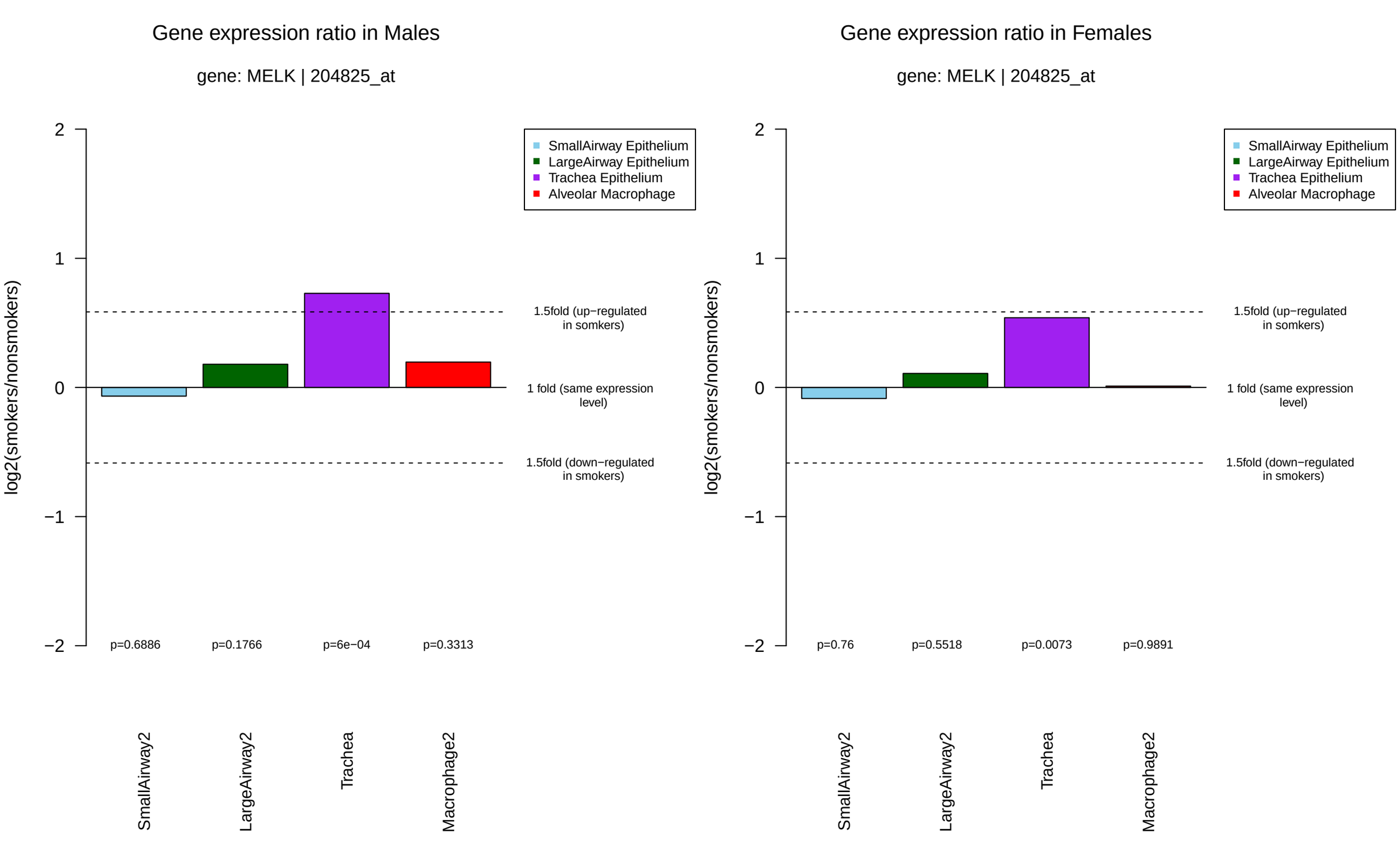
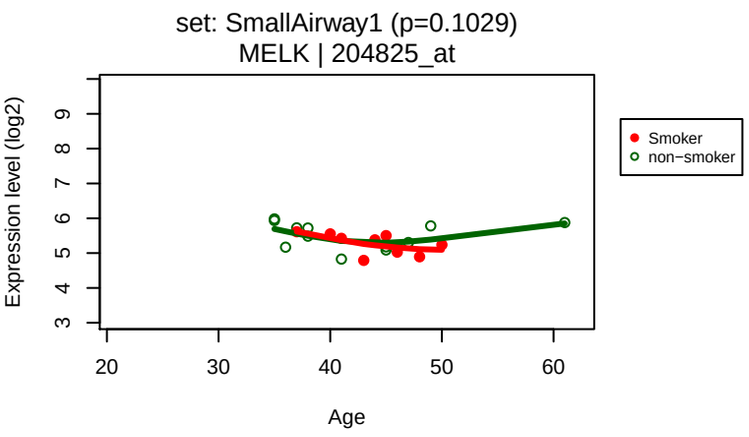
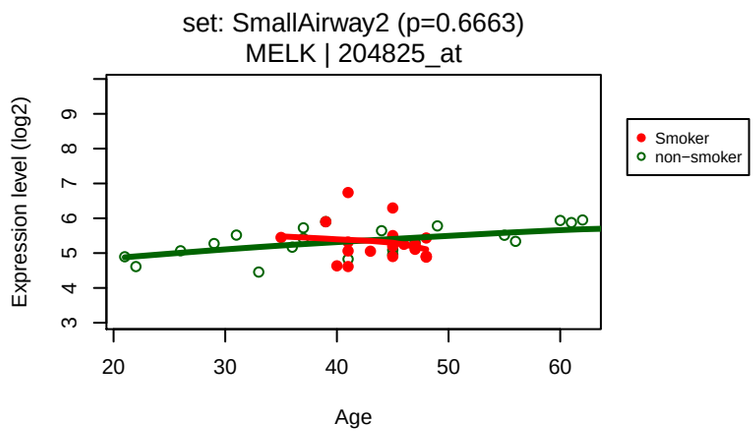
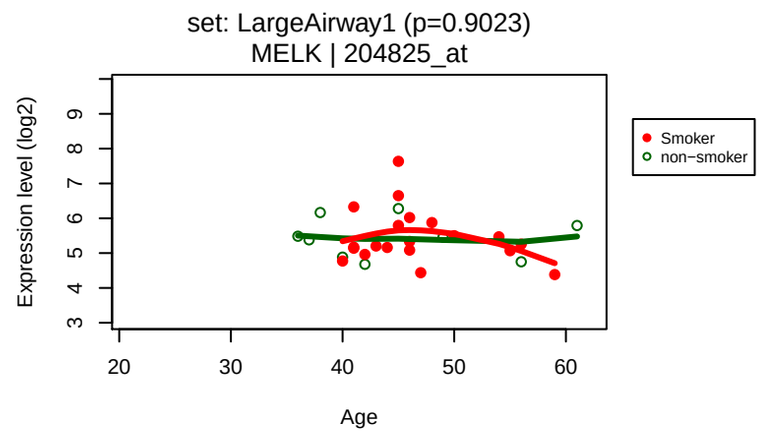

Gene expression of smokers and nonsmokers
gene: MELK | 204825_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1029
p=0.6663
p=0.014
p=0.9023
p=0.0958
p=4e−04
p=0.9912
p=0.5515
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

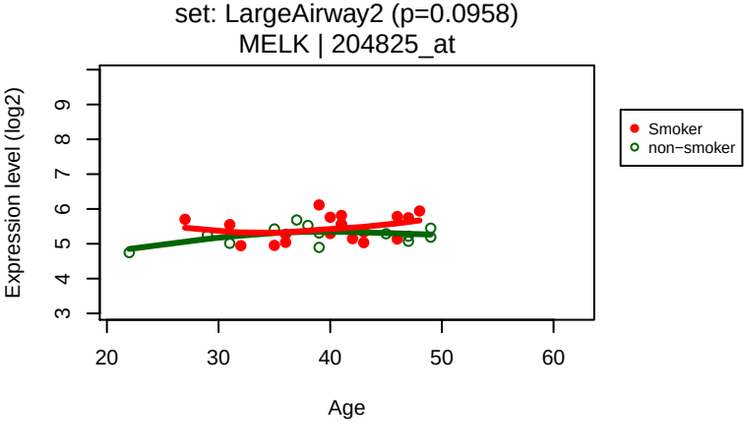

Gene expression ratio (smokers/nonsmokers)
gene: MELK | 204825_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1029
p=0.6663
p=0.014
p=0.9023
p=0.0958
p=4e−04
p=0.9912
p=0.5515
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

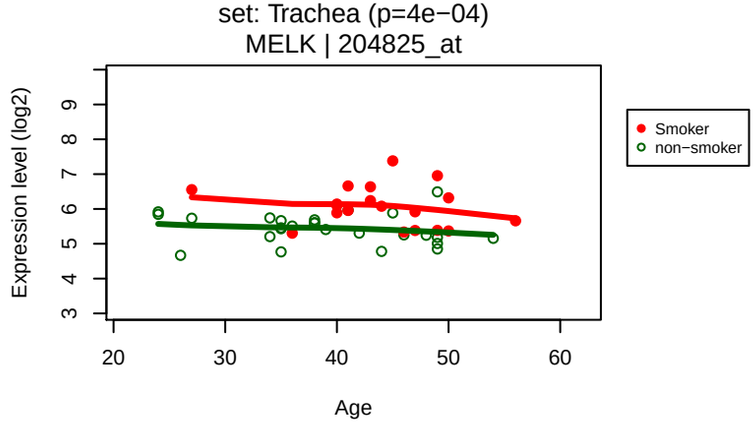

Gene expression ratio in Males
gene: MELK | 204825_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6886
p=0.1766
p=6e−04
p=0.3313
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MELK | 204825_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.76
p=0.5518
p=0.0073
p=0.9891
SmallAirway2
LargeAirway2
Trachea
Macrophage2

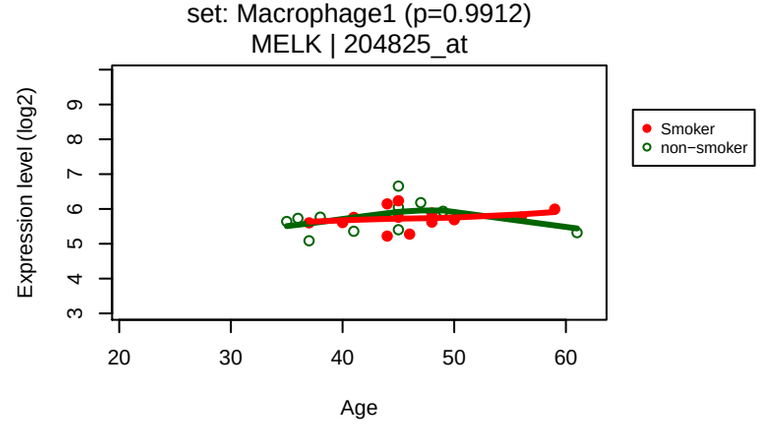

set: SmallAirway1 (p=0.1029)
MELK | 204825_at
Expression level (log2)
Age
Smoker
non−smoker

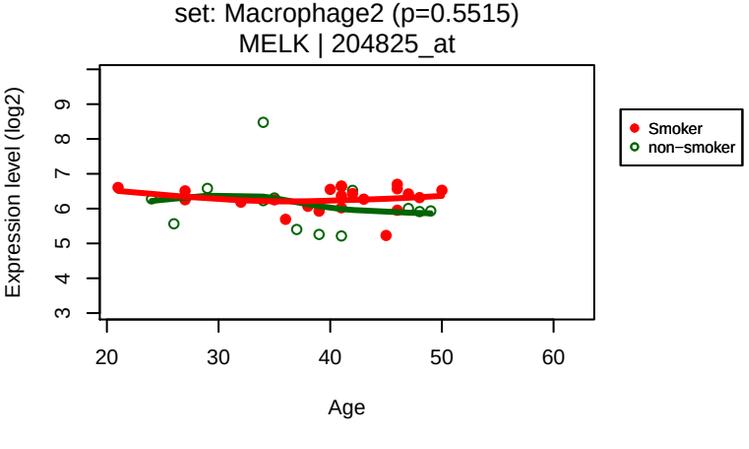

set: SmallAirway2 (p=0.6663)
MELK | 204825_at
Expression level (log2)
Age
Smoker
non−smoker

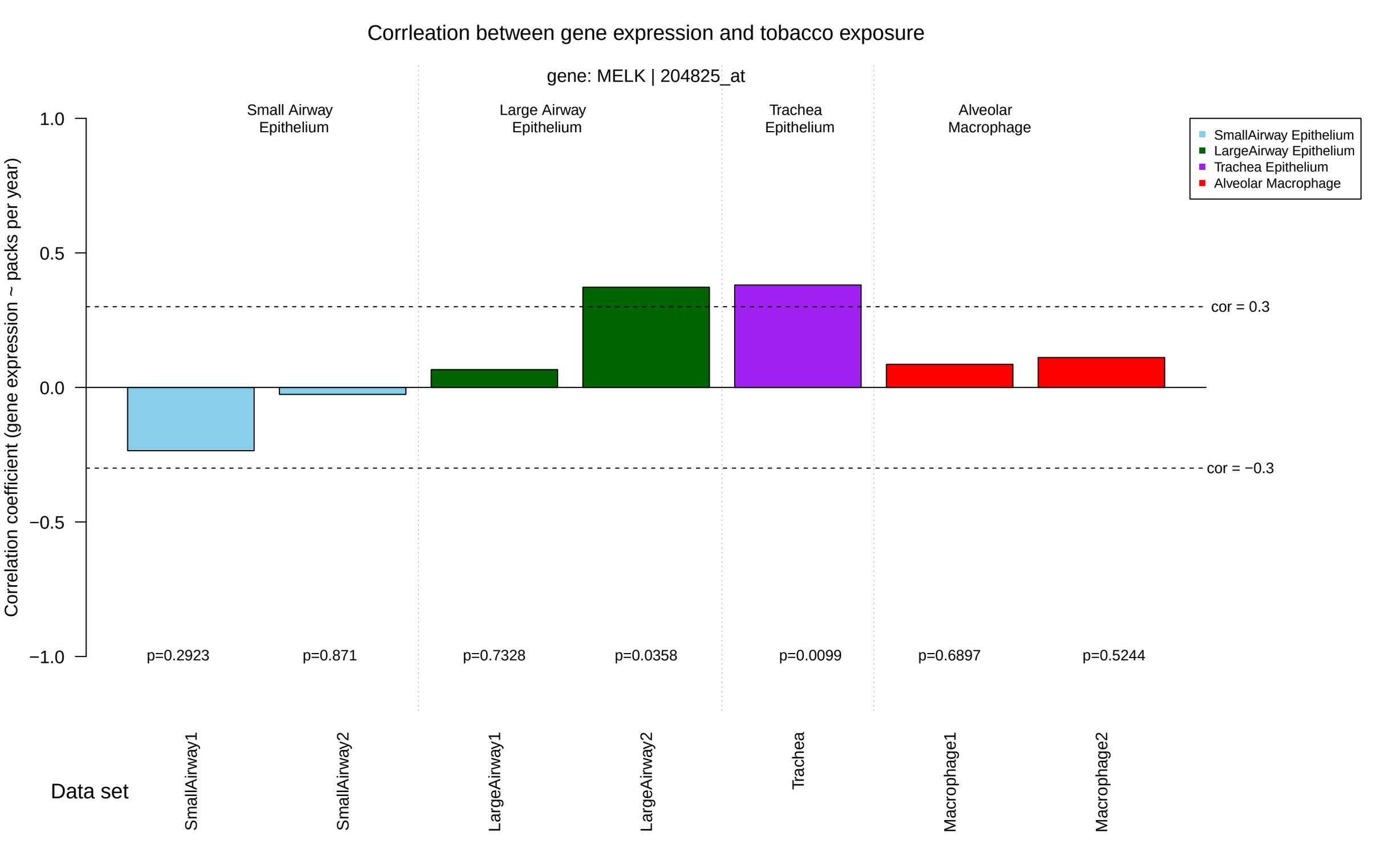

set: LargeAirway1 (p=0.9023)
MELK | 204825_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0958)
MELK | 204825_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=4e−04)
MELK | 204825_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9912)
MELK | 204825_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5515)
MELK | 204825_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MELK | 204825_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2923
p=0.871
p=0.7328
p=0.0358
p=0.0099
p=0.6897
p=0.5244
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2